

Towers

Time limit: 1 s Memory limit: 256 MiB

You have n computers and m towers all at distinct positions along a line. You need to pair computers using cables in such a way that every cable starts at some computer, visits some towers and ends at another computer. The cable can visit any of the towers (not just the ones between the computers) in an arbitrary order. It can skip towers by passing by without visiting them. It can also visit no towers at all and connect the two computers directly but it cannot connect a computer to itself. The number of computers is even.

Let a and b be the positions of two computers and let $x_1, \dots, x_k$ be positions of the towers the cable visits. The length of the cable is $|a - x_1| + |x_1 - x_2| + \dots + |x_{k-1} - x_k| + |x_k - b|$. For some cable we define its score as $f \cdot u - l$, where l is its length, f is some fixed constant and u is the number of unique towers the cable visits among $x_1, \dots, x_k$. Multiple cables can visit the same tower and that tower contributes to the score of each of those cables.

You need to compute the maximum possible sum of cable scores for pairing up all computers (i.e. each computer has to belong to exactly one pair, or equivalently, it must be connected to exactly one cable).

Input

The first line contains the number of test cases, T . The test cases follow one after another. Each test case consists of three lines. The first line contains three integers n , m and f — the number of computers, the number of towers and the constant f . The second line contains n integers $a_1, a_2, \dots, a_n$ — the positions of computers. The third line contains m integers $b_1, b_2, \dots, b_m$ — the positions of towers.

Constraints

Let N and M be the sum of n and m through all test cases, respectively.

- $1 \leq T \leq 10^4$
- $1 \leq N, M \leq 2 \cdot 10^5$
- $0 \leq f \leq 10^9$
- n is even
- $1 \leq a_i, b_i \leq 10^9$
- All positions of computers and towers are unique (within an individual test case).

Subtasks

- Subtask 1 (5 points): $N \leq 5000$, $m = 1$
- Subtask 2 (10 points): $T \leq 20$, $n \leq 10$, $m \leq 100$
- Subtask 3 (27 points): $N, M \leq 5000$
- Subtask 4 (21 points): $N \leq 5000$
- Subtask 5 (37 points): No additional constraints.

Output

Output T integers, each on its own line — the maximum possible sum of scores of cables for each test case.

Example

Input

```
4
2 1 100
1 10
11
4 1 10
2 4 6 8
20
4 1 10
2 4 6 8
5
6 3 10
2 13 4 8 6 10
5 1 9
```

Output

```
89
-4
12
51
```